

GLOBAL INFLATION ANALYSIS (2000–2025)

Using Python and Power BI

Independent Data Analytics Project Report

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Project Focus

Economic Data Analytics, Inflation Analysis, Visualization, and Forecasting

Project Type

Independent Portfolio Research Project

Tools and Technologies

Python • Pandas • NumPy • Matplotlib • Scikit-learn • OpenPyXL • Microsoft Excel •
Power BI

Dataset Source

World Bank Open Data

Indicator: Inflation, Consumer Prices (Annual %)

Indicator Code: FP.CPI.TOTL.ZG

Study Period

2000–2025

Date

August 2026

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Abstract

Inflation is one of the most significant macroeconomic indicators affecting economic growth, purchasing power, investment decisions, and monetary policy. Understanding long-term inflation trends across countries provides valuable insights into global economic stability and the effectiveness of policy responses. This project presents a comprehensive analysis of global inflation trends from 2000 to 2025 using the World Bank's Consumer Price Inflation (Annual %) dataset.

The analysis was conducted using Python for data cleaning, exploratory data analysis, statistical analysis, visualization, and forecasting, while Power BI was used to develop an interactive dashboard for presenting key insights. The dataset was cleaned by removing unnecessary variables, selecting the study period (2000–2025), handling missing values, and eliminating observations with excessive missing data. Several analytical techniques were applied, including trend analysis, country ranking, inflation volatility analysis, regional comparison, correlation analysis, and linear regression forecasting.

The study identified significant fluctuations in global inflation over the study period, with major inflation peaks occurring during the 2008 global financial crisis, the COVID-19 pandemic, and the 2022 global inflation surge driven by supply chain disruptions and geopolitical tensions. The analysis also identified countries with consistently high inflation, countries exhibiting stable inflation performance, and regional differences in inflation patterns. Forecasting results indicated a gradual moderation in global inflation following recent inflationary shocks.

This project demonstrates practical applications of data analytics in economics by integrating statistical analysis, data visualization, predictive modeling, and business intelligence tools. The findings provide a clear understanding of global inflation dynamics while showcasing the use of Python and Power BI to transform large economic datasets into meaningful insights for researchers, policymakers, and decision-makers.

Keywords: Inflation, Consumer Price Index, Global Economy, Data Analytics, Python, Power BI, Forecasting, Economic Research.

1.Introduction

Inflation is a key macroeconomic indicator that measures the rate at which the general price level of goods and services increases over time. As inflation rises, the purchasing power of money declines, affecting households, businesses, investors, and governments. Maintaining stable inflation is one of the primary objectives of central banks because excessive inflation or prolonged deflation can create economic instability, reduce consumer confidence, and hinder sustainable economic growth.

Over the past two decades, the global economy has experienced several major economic events that have significantly influenced inflation patterns. The 2008 Global Financial Crisis, the COVID-19 pandemic, disruptions in global supply chains, rising energy prices, and geopolitical conflicts such as the Russia–Ukraine war have all contributed to substantial changes in inflation rates across countries. These events have highlighted the importance of monitoring inflation trends to better understand economic conditions and support informed policy decisions.

Advances in data analytics have transformed the way economic data is analysed. Programming languages such as Python enable researchers to clean large datasets, perform statistical analysis, create informative visualizations, and develop predictive models efficiently. Similarly, Power BI allows complex economic information to be presented through interactive dashboards, making analytical results more accessible to policymakers, researchers, and business professionals.

This project analyses global consumer price inflation from 2000 to 2025 using data obtained from the World Bank Open Data database. The study applies a complete data analytics workflow, including data cleaning, exploratory data analysis, visualization, inflation ranking, volatility analysis, regional comparison, correlation analysis, and forecasting. Python was used to perform the analytical tasks, while Power BI was used to design an interactive dashboard for presenting the findings.

The objective of this research is not only to examine long-term global inflation trends but also to demonstrate practical applications of data science techniques in economic research. By combining economic theory with modern analytical tools, this project provides meaningful insights into inflation dynamics while showcasing technical competencies in Python, data visualization, business intelligence, and economic data analysis.

2. Problem Statement

Inflation has become one of the most significant economic challenges affecting both developed and developing countries. Although inflation data is publicly available through organizations such as the World Bank, it is often stored in large datasets that are difficult to interpret without proper analysis and visualization. As a result, policymakers, researchers, and students may struggle to identify long-term inflation trends, compare countries, assess inflation stability, or understand regional differences.

Moreover, recent global events—including the COVID-19 pandemic, supply chain disruptions, and geopolitical conflicts—have caused sharp fluctuations in inflation rates worldwide, increasing the need for comprehensive data-driven analysis. Existing raw datasets alone do not provide meaningful insights into these changing patterns.

This project addresses this problem by transforming raw global inflation data into a structured and interactive analytical framework using Python and Power BI. Through data cleaning, exploratory data analysis, visualization, statistical analysis, and forecasting, the study identifies inflation trends, compares country performance, measures inflation stability, examines regional patterns, and predicts future inflation behavior. The project demonstrates how modern data analytics tools can support evidence-based economic research and improve the interpretation of large macroeconomic datasets.

3. Research Objectives

The primary objective of this study is to analyze global inflation trends between **2000 and 2025** using data analytics techniques and interactive visualization tools. The specific objectives of the study are:

1. **To analyze global inflation trends** from 2000 to 2025 using the World Bank Consumer Price Inflation dataset.
2. **To clean and preprocess the raw dataset** by removing unnecessary variables, handling missing values, and preparing the data for analysis.
3. **To identify countries with the highest and lowest average inflation rates** during the study period.

4. **To measure inflation stability** by calculating inflation volatility and identifying the most stable economies.
5. **To compare inflation patterns across different world regions** and examine regional differences in inflation behavior.
6. **To forecast future global inflation trends** using a Linear Regression model based on historical inflation data.
7. **To develop an interactive Power BI dashboard** that presents key inflation indicators and visualizations for effective decision-making.
8. **To demonstrate the practical application of Python and Power BI** in economic data analysis, visualization, and research.

4. Research Questions

This study aims to answer the following research questions:

1. How has global inflation changed between 2000 and 2025?
2. Which countries experienced the highest and lowest average inflation rates during the study period?
3. Which countries maintained the most stable inflation rates over time?
4. How do inflation patterns differ across various world regions?
5. What major global economic events are reflected in the observed inflation trends?
6. Can historical inflation data be used to forecast future global inflation trends using a Linear Regression model?
7. How can Python and Power BI be used to transform raw economic data into meaningful insights through data visualization and interactive dashboards?

5. Dataset Description

5.1 Data Source

The dataset used in this research was obtained from the **World Bank Open Data** database, one of the world's most reliable and widely used sources of international economic statistics. The World Bank collects and publishes macroeconomic indicators for countries across the globe, making it an appropriate source for comparative economic analysis.

Dataset Source: World Bank Open Data

Indicator: Inflation, Consumer Prices (Annual %)

Indicator Code: FP.CPI.TOTL.ZG

5.2 Study Period

The original dataset contained annual inflation data from **1960 to 2025** for countries and regional aggregates worldwide.

To ensure that the analysis focused on contemporary inflation dynamics and major recent economic events, the study selected data covering the period:

2000–2025

This 26-year period captures several significant global economic events, including:

- The Dot-com recession (early 2000s)
- The 2008 Global Financial Crisis
- The European Debt Crisis
- The COVID-19 pandemic
- Global supply chain disruptions
- The 2022 global inflation surge

5.3 Dataset Structure

The original dataset contained:

- **265 countries and regional economies**
- Annual inflation observations from **1960 to 2025**
- Country Name
- Country Code
- Indicator Name
- Indicator Code

After selecting only the required variables and limiting the study period to 2000–2025, the dataset consisted of:

- Country Name
- Country Code
- Annual inflation values (2000–2025)

5.4 Data Cleaning and Preprocessing

Several preprocessing steps were performed before analysis to improve data quality and ensure reliable results.

The following steps were carried out:

- Removed unnecessary columns including *Indicator Name*, *Indicator Code*, and empty columns.
- Selected annual inflation data for the period **2000–2025**.
- Examined missing values across all variables.
- Removed countries with excessive missing observations to improve data reliability.
- Calculated the average inflation rate for each country.
- Calculated inflation volatility using the standard deviation of annual inflation rates.
- Created a regional classification for comparative regional analysis.
- Prepared the cleaned dataset for visualization, forecasting, and dashboard development.

5.5 Missing Value Analysis

The dataset contained missing observations because inflation data were unavailable for some countries during certain years.

Before preprocessing:

- Total observations: **265 countries**
- Total missing values: **985**

To improve analytical reliability, countries with excessive missing observations were removed.

After preprocessing:

- **226 countries** remained for analysis.

This ensured that statistical analysis and forecasting were based on sufficiently complete datasets while maintaining broad global coverage.

5.6 Final Dataset

The final analytical dataset included:

Feature	Description
Countries	226
Study Period	2000–2025
Time Frequency	Annual
Indicator	Consumer Price Inflation (Annual %)
Data Source	World Bank Open Data
Programming Language	Python
Visualization Tool	Power BI

6. Methodology

6.1 Research Design

This study follows a **quantitative research design** using secondary macroeconomic data obtained from the World Bank Open Data database. A data analytics approach was adopted to examine global inflation trends through statistical analysis, data visualization, predictive modeling, and business intelligence techniques. The research combines economic analysis with modern programming tools to generate meaningful insights from large-scale economic data.

6.2 Data Collection

The dataset was collected from the **World Bank Open Data** platform. The selected indicator was **Consumer Price Inflation (Annual %)**, which measures the annual

percentage change in the Consumer Price Index (CPI). The original dataset contained annual observations from **1960 to 2025** for **265 countries and regional economies**.

6.3 Data Preprocessing

Before conducting the analysis, the dataset underwent several preprocessing steps to improve its quality and usability. Unnecessary variables, including *Indicator Name*, *Indicator Code*, and empty columns, were removed. The study period was restricted to **2000–2025** to focus on recent global inflation trends. Missing values were examined, and countries with excessive missing observations were excluded. The cleaned dataset was then prepared for statistical analysis and visualization.

6.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand inflation behavior across countries and over time. Summary statistics and multiple visualizations were created to identify long-term trends, compare country performance, evaluate inflation stability, and detect unusual patterns. This stage provided the foundation for interpreting the dataset before predictive analysis.

6.5 Statistical Analysis

Several statistical measures were calculated to support the analysis. The average inflation rate was computed for each country to identify countries with the highest and lowest inflation. Inflation volatility was measured using the standard deviation of annual inflation rates to evaluate economic stability. Regional averages were also calculated to compare inflation behavior across different parts of the world.

6.6 Predictive Modeling

A **Linear Regression** model was developed using historical global average inflation data to forecast future inflation trends. The model estimated the relationship between time (years) and inflation rates, providing projected inflation values beyond the available dataset. Although Linear Regression is a simple forecasting technique, it effectively demonstrates the application of predictive analytics in economic research.

6.7 Data Visualization

Data visualization played an essential role in communicating the analytical findings. Multiple charts and graphs were developed using **Matplotlib**, including:

- Global Inflation Trend
- Top 10 Highest Inflation Countries
- Top 10 Lowest Inflation Countries
- Pakistan Inflation Trend
- Inflation Volatility
- Most Stable Inflation Countries
- Regional Inflation Comparison
- Correlation Heatmap
- Inflation Forecast

These visualizations transformed raw numerical data into clear and interpretable insights.

6.8 Dashboard Development

An interactive dashboard was created in **Power BI** using the cleaned dataset generated in Python. The dashboard includes Key Performance Indicators (KPIs), trend analysis, country comparisons, regional analysis, and interactive filters, enabling users to explore inflation patterns dynamically. This integration of Python and Power BI demonstrates a complete data analytics workflow from data preparation to business intelligence reporting.

6.9 Software and Tools

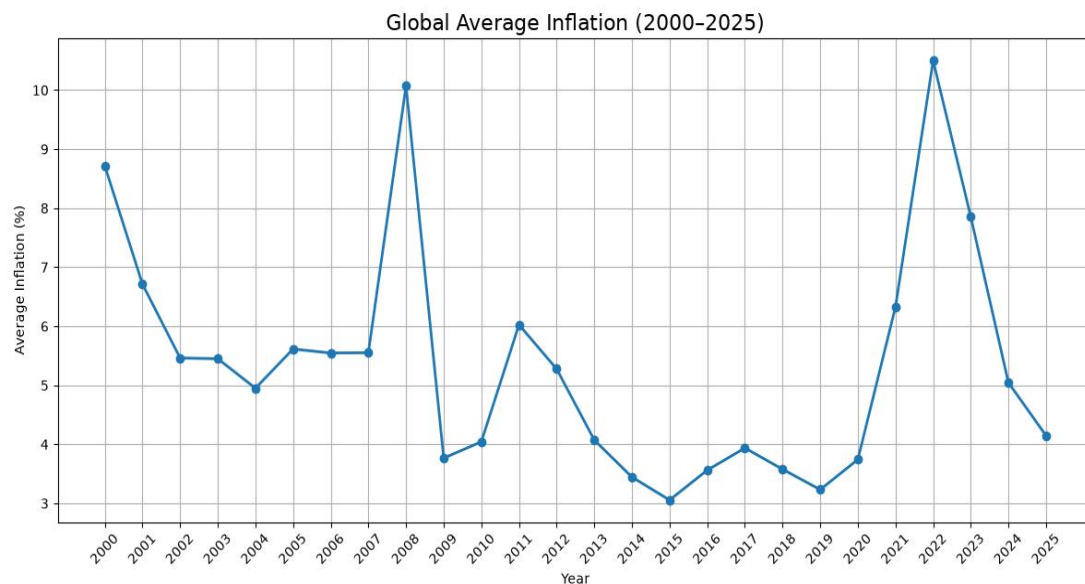
The analysis was conducted using the following software and libraries:

Tools	Purpose
Python	Data Analysis and Programming
Pandas	Data cleaning and manipulation
NumPy	Numerical computations
Matplotlib	Data visualization
Scikit-learn	Linear Regression forecasting
Open PyXL	Excel file generation
Power BI	Interactive Dashboard development
Microsoft Excel	Data storage and verification

7. Exploratory Data Analysis (EDA)

7.1 Global Inflation Trend (2000–2025)

Figure 7.1: Global Inflation Trend (2000–2025)



Objective

To examine the long-term trend of global inflation from 2000 to 2025 and identify periods of significant inflationary changes.

Interpretation

Movement of global average inflation over the study period. Inflation remained relatively moderate during the early 2000s before increasing sharply during the **2008 Global Financial Crisis**. A noticeable decline followed in 2009 as economic activity slowed worldwide.

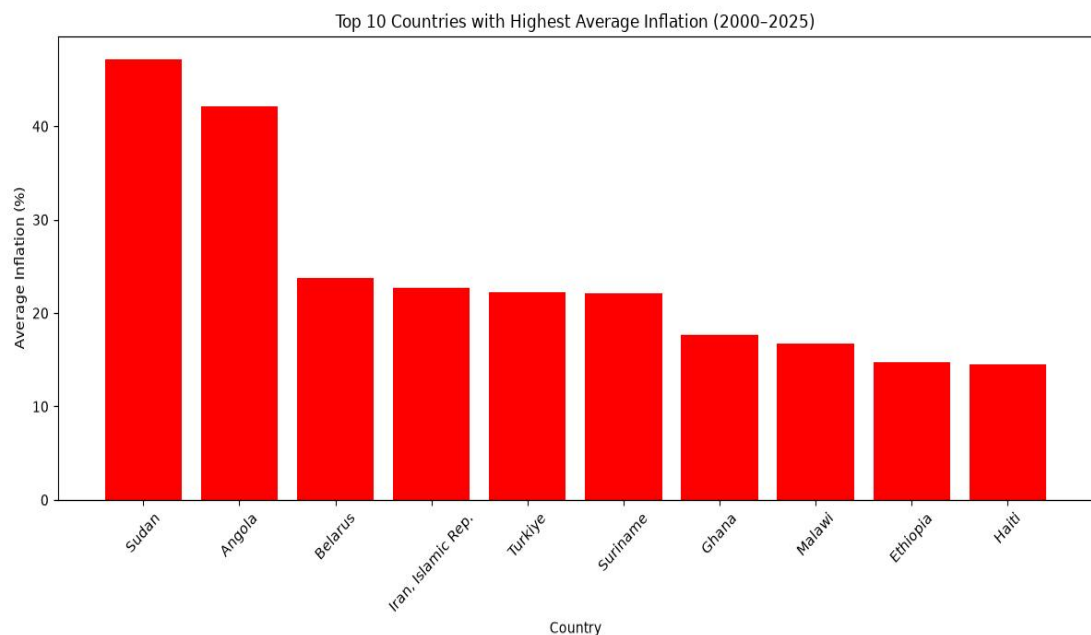
Inflation remained relatively stable between **2013 and 2019**, after which another substantial increase occurred during the **COVID-19 pandemic** and the subsequent global supply chain disruptions. The highest global average inflation was observed in **2022**, reflecting worldwide economic pressures caused by rising energy prices, food shortages, and geopolitical conflicts. Following this peak, inflation gradually declined in **2023–2025**, indicating the effects of monetary tightening and stabilization policies implemented by central banks.

Key Findings

- Global inflation generally followed major global economic events.
- The highest average inflation occurred in **2022**.
- Inflation declined after 2022 but remained above pre-pandemic levels.
- Major crises significantly influenced inflation across countries.

7.2 Top 10 Countries with Highest Average Inflation

Figure 7.2: Top 10 Countries with Highest Average Inflation



Objective

To identify the countries that experienced the highest average annual inflation during the study period (2000–2025) and compare their inflation performance.

Interpretation

Ten countries with the highest average inflation rates over the study period. The results indicate that **Sudan** recorded the highest average inflation, followed by **Angola, Belarus, Iran, Türkiye, Suriname, Ghana, Malawi, Ethiopia, and Haiti**.

These countries experienced prolonged periods of high inflation due to a combination of economic instability, currency depreciation, political uncertainty, fiscal imbalances, and external economic shocks. High inflation in these economies reduced purchasing power, increased the cost of living, and created challenges for sustainable economic growth.

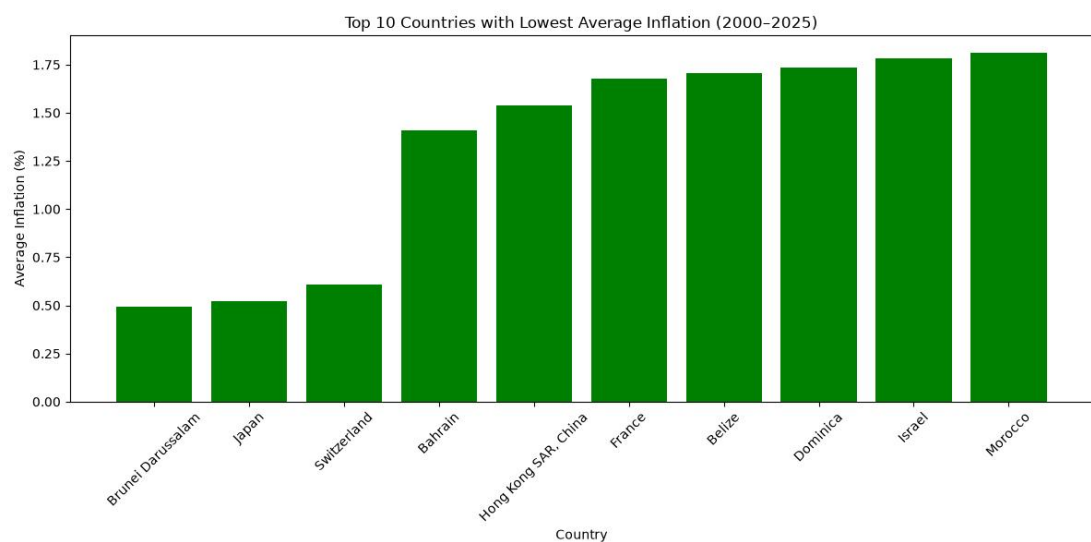
The analysis demonstrates that inflation is not distributed uniformly across countries and that developing economies are generally more vulnerable to persistent inflationary pressures than advanced economies.

Key Findings

- Sudan recorded the highest average inflation during 2000–2025.
- Angola and Belarus also experienced persistently high inflation.
- Most high-inflation countries are developing economies.
- Economic and political instability were major contributors to sustained inflation.

7.3 Top 10 Countries with Lowest Average Inflation

Figure 7.3: Top 10 Countries with Lowest Average Inflation



Objective

To identify the countries that maintained the lowest average inflation rates during the study period and evaluate their price stability.

Interpretation

Ten countries with the lowest average inflation rates between 2000 and 2025. These countries maintained relatively stable price levels throughout the study period, reflecting effective monetary policies, sound macroeconomic management, and stable economic conditions.

Low and stable inflation contributes to sustainable economic growth by preserving purchasing power, encouraging investment, and reducing uncertainty for businesses and consumers. Countries with consistently low inflation are generally considered to have stronger monetary institutions and more effective inflation-targeting policies.

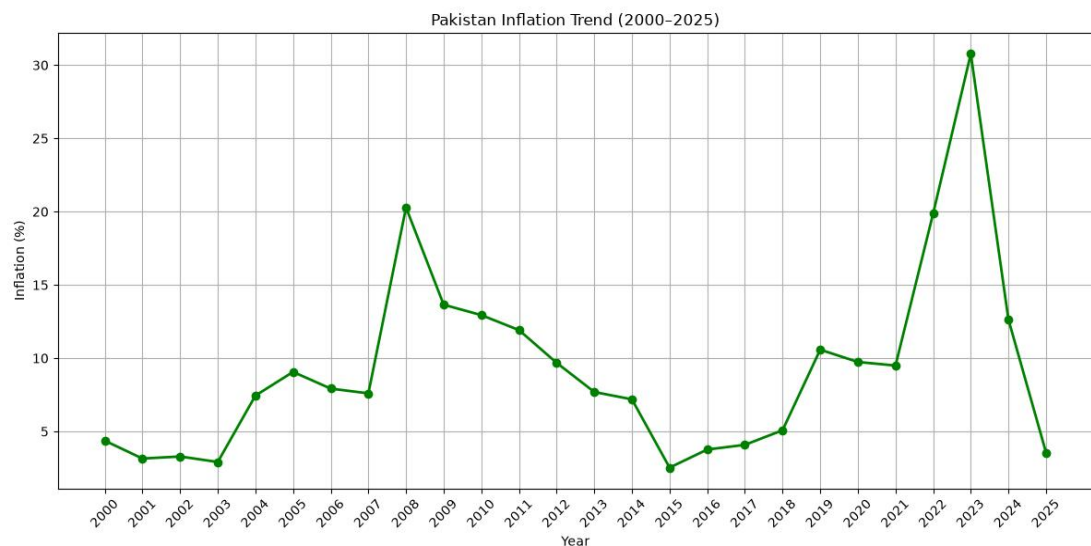
The findings highlight the importance of maintaining price stability as a key objective of economic policy.

Key Findings

- The selected countries recorded the lowest average inflation during 2000–2025.
- Stable inflation indicates stronger macroeconomic performance.
- Effective monetary policy plays an important role in controlling inflation.
- Low inflation supports long-term economic growth and investment.

7.4 Pakistan Inflation Trend (2000–2025)

Figure 7.4: Pakistan Inflation Trend (2000–2025)



Objective

To examine the trend of inflation in Pakistan between 2000 and 2025 and identify periods of significant inflationary changes.

Interpretation

Pakistan's annual inflation trend over the study period. Inflation remained relatively moderate during the early 2000s before increasing significantly during the **2008 Global Financial Crisis**, reflecting rising global commodity prices and domestic economic challenges.

Following a period of relative stability between **2014 and 2019**, inflation accelerated sharply after **2020** due to the economic effects of the COVID-19 pandemic, exchange rate depreciation, supply chain disruptions, rising energy prices, and increasing import costs. Inflation reached one of its highest levels during **2022–2023**, placing considerable pressure on household purchasing power and business activity.

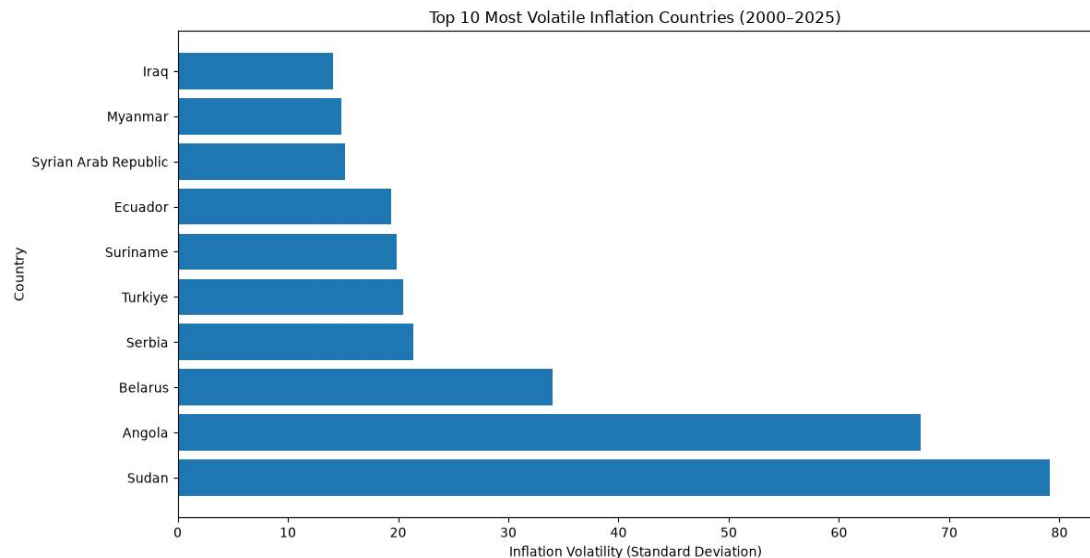
Although inflation began to decline in the later years of the study period, it remained above historical averages, indicating that inflationary pressures continued to affect Pakistan's economy.

Key Findings

- Pakistan experienced several periods of high inflation during 2000–2025.
- Inflation increased sharply during the 2008 financial crisis and again after the COVID-19 pandemic.
- The highest inflation was observed during 2022–2023.
- External economic shocks and domestic macroeconomic challenges significantly influenced Pakistan's inflation trend.

7.5 Inflation Volatility Analysis

Figure 7.5: Inflation Volatility Across Countries



Objective

To evaluate the variability of inflation across countries by measuring the standard deviation of annual inflation rates between 2000 and 2025.

Interpretation

The level of inflation volatility among the countries included in the study. Inflation volatility measures the extent to which inflation rates fluctuate over time. Countries with higher volatility experienced frequent and substantial changes in inflation, indicating greater economic uncertainty and unstable price levels. In contrast, countries with lower volatility maintained relatively stable inflation throughout the study period.

High inflation volatility often reflects macroeconomic instability, exchange rate fluctuations, fiscal imbalances, political uncertainty, or external economic shocks. Conversely, lower volatility suggests stronger monetary policy frameworks and more effective inflation management.

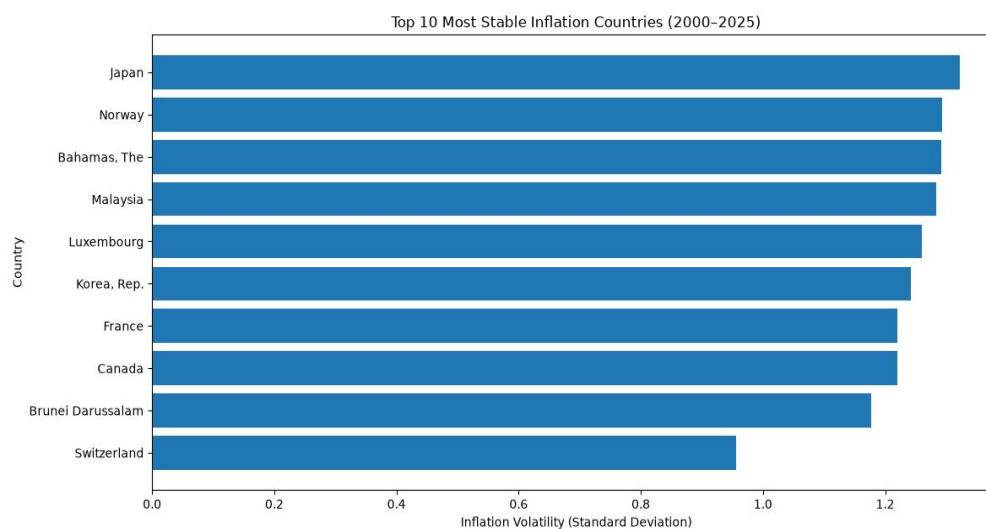
The analysis highlights that inflation stability is as important as maintaining a low average inflation rate, as predictable inflation supports investment, economic planning, and sustainable growth.

Key Findings

- Inflation volatility varied considerably across countries.
- Economies experiencing political or economic instability generally showed higher inflation volatility.
- Stable economies recorded lower fluctuations in annual inflation rates.
- Inflation stability is an important indicator of overall macroeconomic performance.

7.6 Top 10 Most Stable Inflation Countries

Figure 7.6: Top 10 Most Stable Inflation Countries



Objective

To identify the countries that maintained the most stable inflation rates between 2000 and 2025 by analyzing inflation volatility.

Interpretation

Ten countries with the lowest inflation volatility during the study period. Countries such as **Switzerland, Brunei Darussalam, Canada, France, South Korea, Luxembourg, Malaysia, The Bahamas, Norway, and Japan** consistently maintained stable inflation, reflecting effective monetary policies and strong macroeconomic management.

Stable inflation is a key indicator of economic health because it reduces uncertainty for consumers, businesses, and investors. Countries with low inflation volatility are generally able to preserve purchasing power, encourage long-term investment, and maintain sustainable economic growth.

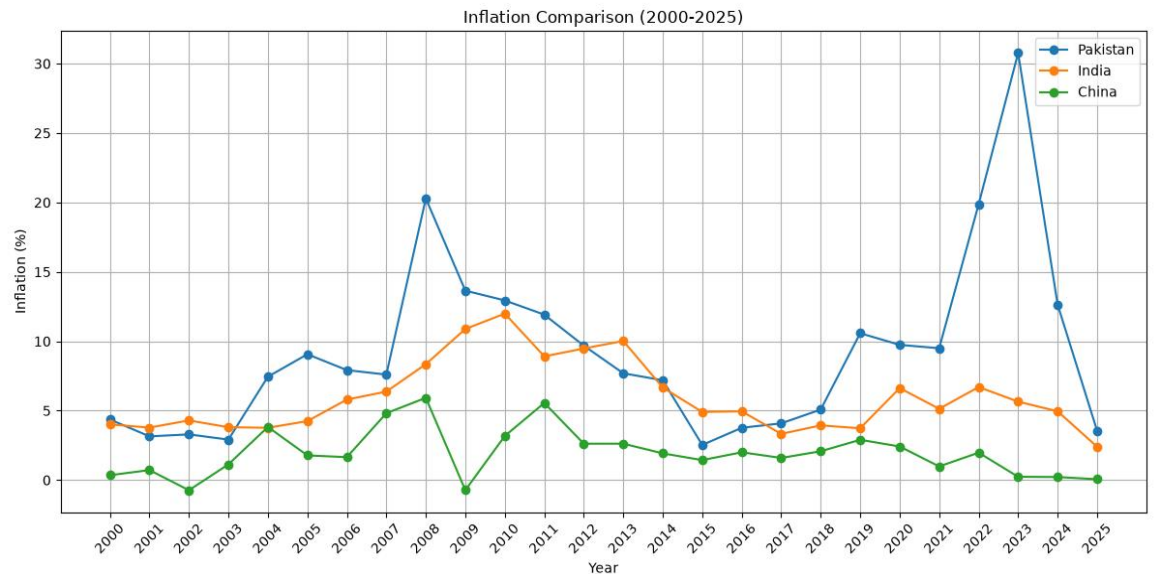
The findings demonstrate that advanced and well-managed economies tend to experience more predictable inflation patterns due to disciplined fiscal policies, independent central banks, and effective inflation-targeting frameworks.

Key Findings

- Switzerland recorded the lowest inflation volatility during the study period.
- Most of the most stable countries are developed or high-income economies.
- Effective monetary policy contributes significantly to maintaining stable inflation.
- Inflation stability supports economic confidence, investment, and long-term growth.

7.7 Inflation Comparison: Pakistan, India, and China

Figure 7.7: Inflation Comparison of Pakistan, India, and China (2000–2025)



Objective

To compare inflation trends in Pakistan, India, and China between 2000 and 2025 and examine differences in their inflation experiences.

Interpretation

The comparison of Pakistan, India, and China shows differences in inflation patterns over the study period. Pakistan experienced relatively greater fluctuations and higher inflationary pressures during several periods, while India and China generally followed comparatively different and more moderate inflation patterns.

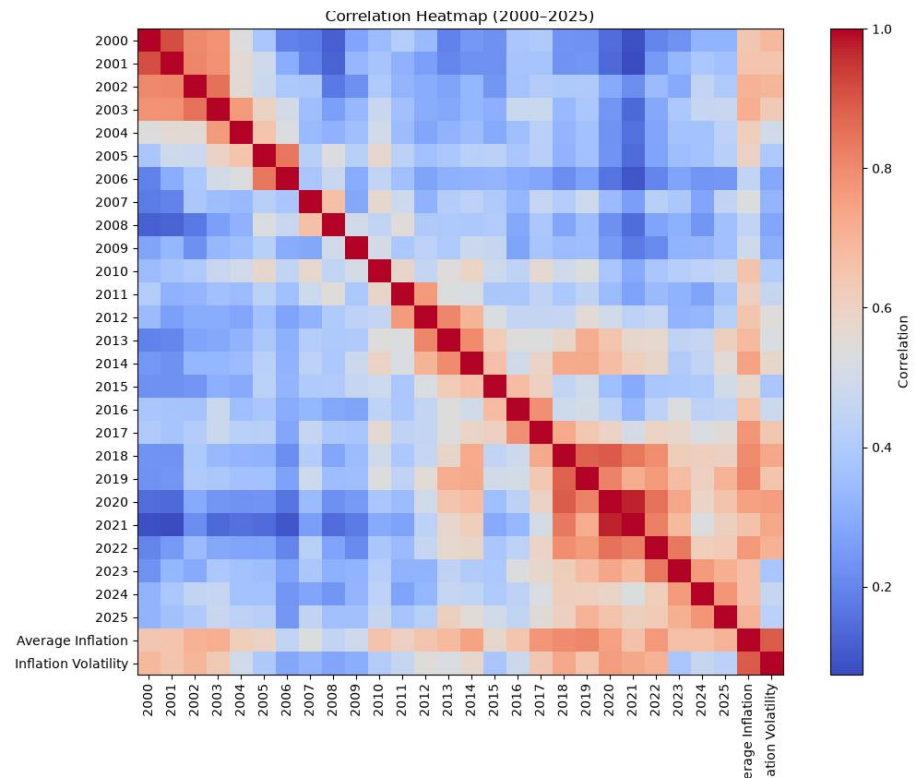
The differences reflect variations in domestic economic conditions, monetary and fiscal policies, exchange-rate movements, supply-side pressures, and exposure to external economic shocks.

Key Findings

- Pakistan experienced comparatively higher inflation volatility during several periods.
- India maintained a relatively more moderate inflation pattern over much of the study period.
- China generally experienced lower and more stable inflation compared with Pakistan.
- The comparison demonstrates tha

7.8 Correlation Heatmap Analysis

Figure 7.8: Correlation Heatmap Analysis of Annual Inflation Rates



Objective

To examine the correlation between annual inflation rates from 2000 to 2025 and identify how inflation patterns changed over time.

Interpretation

The correlation matrix of annual global inflation rates across the study period. The heatmap illustrates the strength and direction of relationships between inflation rates in different years. Darker colors indicate stronger positive correlations, while lighter colors represent weaker relationships.

The analysis shows that inflation rates in consecutive years generally exhibit positive correlations, suggesting that inflation tends to persist over time rather than changing abruptly. Strong correlations were observed among recent years, particularly during

periods affected by common global economic events such as the COVID-19 pandemic and the subsequent inflation surge.

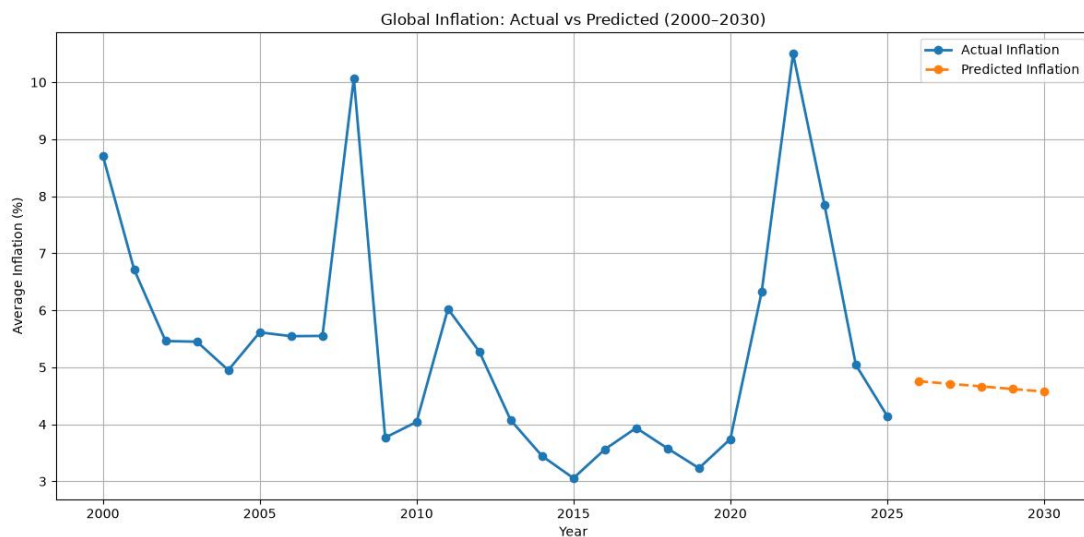
Lower correlations between years that were far apart indicate that inflation dynamics evolve over time due to changes in global economic conditions, monetary policies, and external shocks.

Key Findings

- Most consecutive years show positive correlations in inflation.
- Recent years display stronger correlations because of shared global economic events.
- Inflation patterns evolve over time rather than remaining constant.
- The heatmap confirms the presence of long-term inflation trends across the study period.

7.9 Average Vs Predicted Global Inflation

Figure 7.9: Average vs Predicted Global Inflation



Objective

To compare historical average global inflation with the inflation values predicted by the Linear Regression model and evaluate future inflation trends.

Interpretation

Comparison of the historical average global inflation rates with the values predicted by the Linear Regression model. The blue line represents the observed average inflation over the study period, while the forecast line illustrates the projected long-term trend based on historical data.

The model captures the overall direction of inflation rather than short-term fluctuations. While actual inflation varies due to global economic shocks, the predicted values indicate a gradual stabilization of inflation following the significant rise observed during the post-pandemic period. This comparison demonstrates how historical economic data can be used to estimate future inflation patterns through predictive analytics.

Key Findings

- The model successfully follows the long-term trend of historical inflation.
- Predicted inflation suggests gradual stabilization in future years.
- The comparison highlights the difference between observed and forecasted values.
- Linear Regression provides a simple but effective forecasting approach for economic data.

8. Findings and Discussion

This chapter summarizes the major findings obtained from the statistical analysis, visualizations, and predictive modeling. The results provide valuable insights into global inflation behavior between 2000 and 2025 and demonstrate the effectiveness of Python and Power BI in economic data analysis.

• Major Findings

The analysis revealed several important findings:

1. Global inflation followed major economic events.

The global inflation trend showed noticeable increases during periods of worldwide economic uncertainty, including the **2008 Global Financial Crisis**, the **COVID-19 pandemic**, and the **2022 global inflation surge**. These events significantly influenced inflation across both developed and developing economies.

2. Large differences exist between countries.

The study found considerable variation in inflation among countries. Nations such as **Sudan** and **Angola** experienced persistently high inflation, whereas countries including **Switzerland**, **Canada**, and **Japan** maintained relatively stable price levels throughout the study period.

3. Inflation stability is as important as average inflation.

The volatility analysis demonstrated that countries with low inflation volatility generally had stronger monetary policy frameworks and more stable macroeconomic environments. Stable inflation supports investment, business confidence, and sustainable economic growth.

4. Regional inflation patterns differ significantly.

Regional comparison showed that developing regions generally experienced higher inflation than developed regions. These differences were largely influenced by economic development, fiscal policy, exchange rate stability, and institutional strength.

5. Pakistan experienced significant inflationary pressures.

Pakistan's inflation trend reflected both global and domestic economic challenges. Inflation increased sharply during international crises and remained elevated during the post-pandemic period, highlighting the importance of effective monetary and fiscal policies.

6. Historical data can support inflation forecasting.

The Linear Regression model successfully captured the overall long-term trend in global inflation. Although it cannot predict unexpected economic shocks, it

demonstrated how historical macroeconomic data can be used to estimate future inflation patterns.

7. Python and Power BI provide an effective analytical workflow.

The integration of Python for data cleaning, statistical analysis, visualization, and forecasting with Power BI for dashboard development demonstrated a complete end-to-end data analytics process suitable for economic research and decision-making.

Discussion

The findings suggest that inflation is influenced by a combination of global economic events, domestic policy decisions, and structural economic conditions. Countries with sound monetary institutions and disciplined fiscal policies generally maintained more stable inflation, while economies facing political instability, exchange rate depreciation, or external shocks experienced greater inflationary pressures.

The study also demonstrates the value of modern data analytics tools in economics. By combining Python and Power BI, large macroeconomic datasets can be transformed into meaningful insights that support research, policy analysis, and informed decision-making.

9. Conclusion

This research analyzed global inflation trends between **2000 and 2025** using the World Bank Consumer Price Inflation dataset. By applying data analytics techniques in Python and developing an interactive Power BI dashboard, the study successfully transformed raw macroeconomic data into meaningful economic insights.

The analysis revealed that global inflation has been significantly influenced by major international economic events, including the **2008 Global Financial Crisis**, the **COVID-19 pandemic**, and the **2022 global inflation surge**. Considerable differences in inflation performance were observed across countries and regions, highlighting the role of economic development, monetary policy, and political stability in determining inflation outcomes.

The study also identified countries with the highest and lowest average inflation rates, measured inflation volatility to assess economic stability, compared regional inflation patterns, and examined Pakistan's inflation trend over the study period. Furthermore, a Linear Regression model demonstrated the practical application of predictive analytics by forecasting future inflation trends based on historical data.

The integration of **Python** and **Power BI** proved to be an effective approach for economic data analysis, enabling efficient data preprocessing, statistical analysis, visualization, forecasting, and dashboard development. The project demonstrates how modern analytical tools can support economic research and provide valuable insights for researchers, policymakers, and decision-makers.

Overall, this study achieved its research objectives and illustrates the growing importance of data science and business intelligence techniques in the field of economics.

10. Recommendations

Based on the findings of this study, the following recommendations are proposed:

1. Strengthen Monetary Policy

Central banks should continue implementing effective monetary policies to maintain price stability and control inflation. Timely adjustments in interest rates and inflation-targeting strategies can help reduce inflationary pressures.

2. Improve Fiscal Discipline

Governments should adopt responsible fiscal policies by controlling budget deficits, reducing unnecessary public expenditure, and improving tax collection systems. Sound fiscal management contributes to long-term economic stability.

3. Enhance Supply Chain Resilience

Since global inflation has been strongly influenced by supply chain disruptions, governments should invest in logistics, transportation infrastructure, and domestic production capacity to reduce vulnerability to external shocks.

4. Promote Economic Stability

Countries experiencing persistently high inflation should implement structural economic reforms that improve exchange rate stability, strengthen financial institutions, and encourage productive investment.

5. Expand the Use of Data Analytics

Government agencies, researchers, and policymakers should increasingly utilize data analytics tools such as Python, Power BI, and machine learning techniques to monitor inflation trends and support evidence-based economic decision-making.

6. Develop More Advanced Forecasting Models

Future studies should explore more sophisticated forecasting techniques, including ARIMA, Prophet, Random Forest, and Long Short-Term Memory (LSTM) neural networks, to improve prediction accuracy.

7. Encourage Open Economic Data

International organizations and governments should continue promoting open-access economic datasets, enabling researchers to conduct transparent, reproducible, and high-quality economic research.

11. Limitations of the Study

Every research study has certain limitations that should be considered when interpreting the results. The following limitations were identified in this research:

1. Dependence on Secondary Data

This study relied entirely on secondary data obtained from the World Bank Open Data database. The accuracy and completeness of the analysis therefore depend on the quality of the published data.

2. Missing Observations

Several countries contained missing inflation values for certain years. Although countries with excessive missing data were removed during preprocessing, some missing observations remained, which may slightly affect the analysis.

3. Limited Forecasting Technique

The forecasting analysis used a **Linear Regression** model. While suitable for demonstrating predictive analytics, this model assumes a linear relationship and cannot fully capture complex economic dynamics or unexpected global events.

4. Single Economic Indicator

The study focused only on **Consumer Price Inflation (Annual %)**. Other macroeconomic variables such as GDP growth, unemployment, exchange rates, interest rates, and money supply were not included in the analysis.

5. External Economic Shocks

Unexpected events such as financial crises, pandemics, geopolitical conflicts, and natural disasters can significantly influence inflation. Such events cannot be predicted accurately by simple statistical forecasting models.

6. Regional Classification

Countries were grouped into broad geographical regions for comparison. These classifications may not fully capture differences in economic structures, policy frameworks, or levels of development within each region.

12. Future Research

This study provides a foundation for further research on global inflation using data analytics and visualization techniques. Future studies may expand upon this work in several ways.

1. Incorporate Additional Macroeconomic Variables

Future research can include variables such as Gross Domestic Product (GDP), unemployment rates, exchange rates, interest rates, money supply, and government expenditure to better understand the determinants of inflation.

2. Apply Advanced Machine Learning Models

More sophisticated forecasting models, including **ARIMA**, **Prophet**, **Random Forest**, **XGBoost**, and **Long Short-Term Memory (LSTM)** neural networks, can be used to improve forecasting accuracy and capture complex inflation patterns.

3. Conduct Country-Specific Studies

Researchers may focus on individual countries or regions to investigate the economic, political, and institutional factors influencing inflation in greater detail.

4. Extend the Time Horizon

As new economic data become available, future studies can extend the analysis beyond 2025 to evaluate recent inflation developments and improve long-term forecasting.

5. Comparative Policy Analysis

Future research may compare the effectiveness of different monetary and fiscal policy strategies across countries to identify best practices for maintaining price stability.

6. Develop Interactive Decision-Support Systems

The integration of advanced dashboards, real-time data, and artificial intelligence can provide policymakers with dynamic tools for monitoring inflation and supporting economic decision-making.

13. References

World Bank. (2025). *World Development Indicators: Inflation, consumer prices (annual %)*. <https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG>

Python

Python Software Foundation. (2025). *Python*. <https://www.python.org/>

Pandas

The pandas development team. (2025). *pandas documentation*. <https://pandas.pydata.org/docs/>

NumPy

NumPy Developers. (2025). *NumPy documentation*. <https://numpy.org/doc/>

Matplotlib

Matplotlib Development Team. (2025). *Matplotlib documentation*. <https://matplotlib.org/stable/>

Scikit-learn

Scikit-learn developers. (2025). *Scikit-learn: Machine learning in Python*. <https://scikit-learn.org/stable/>

Microsoft Power BI

Microsoft. (2025). *Power BI documentation*. <https://learn.microsoft.com/power-bi/>

14. Appendices

Appendix A – Python Source Code

The complete Python program developed for this project is provided in this appendix.

The code includes:

- Data loading
- Data cleaning
- Missing value analysis
- Exploratory Data Analysis
- Statistical analysis
- Forecasting using Linear Regression
- Data visualization
- Exporting cleaned datasets

The complete Python source code developed for this project is available in the project's GitHub repository. The script includes data loading, data cleaning, missing-value analysis, exploratory data analysis, statistical analysis, data visualization, and Linear Regression forecasting.



Python File: analysis final.py

GitHub Repository:

Appendix B – Power BI Dashboard

The Power BI dashboard developed for this project provides an interactive visual summary of global inflation trends between 2000 and 2025. The dashboard integrates the cleaned World Bank inflation dataset and presents key indicators and comparisons through interactive visualizations.

The dashboard includes:

- Global inflation trend
- Average, highest, and lowest inflation indicators
- Country-level inflation comparisons

- Regional/country comparison visualizations
- Interactive geographic visualization
- Year and region filters

The dashboard complements the Python-based analysis by providing an interactive platform for exploring the results and identifying inflation patterns across countries and years.

Global Inflation Analysis Dashboard (2000–2025)

Average Inflation (%)

5.36

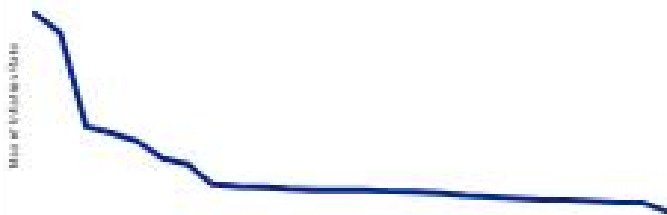
Highest Inflation (%)

359.09

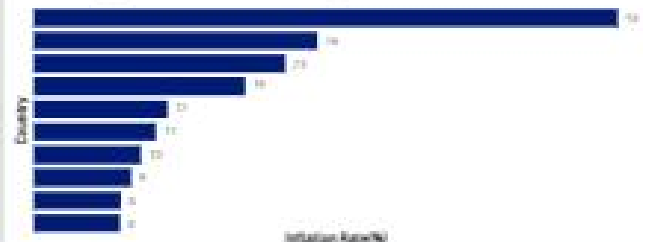
Lowest Inflation (%)

-16.86

Global Inflation Trend



Top 10 Countries By inflation Rate(%)



Select Country

Country Name

all

Select Year

Year

all

Global Inflation by Country



Appendix C – Research Figures

This appendix provides supplementary access to the figures generated during the Python-based analysis. The principal figures and their corresponding interpretations are presented in Chapter 7 of this report. The original high-resolution figures are also available in the project's GitHub repository

.Include all graphs generated during the analysis:

- Figure 7.1 Global Inflation Trend
- Figure 7.2 Highest Inflation Countries
- Figure 7.3 Lowest Inflation Countries
- Figure 7.4 Pakistan Inflation Trend
- Figure 7.5 Inflation Volatility
- Figure 7.6 Most Stable Inflation Countries
- Figure 7.7 Inflation Comparison of Pakistan, India, and China (2000–2025)
- Figure 7.8 Correlation Heatmap
- Figure 7.9 Average vs Predicted Global Inflation

Appendix D – Dataset Information

The dataset used in this project was obtained from the World Bank Open Data database. The analysis focuses on the Consumer Price Index (CPI) inflation indicator, reported as annual percentage change in consumer prices.

Dataset Source: World Bank Open Data

Indicator: Inflation, Consumer Prices (Annual %)

Indicator Code: FP.CPI.TOTL.ZG

Study Period: 2000–2025

Unit of Measurement: Percentage (%)

Data Type: Annual country-level inflation data

The original dataset was processed using Python. Data cleaning included selecting the study period, removing unnecessary columns, examining missing observations, and preparing the dataset for statistical analysis and visualization

Source: World Bank Open Data, *Inflation, consumer prices (annual %)*, Indicator Code FP.CPI.TOTL.ZG.